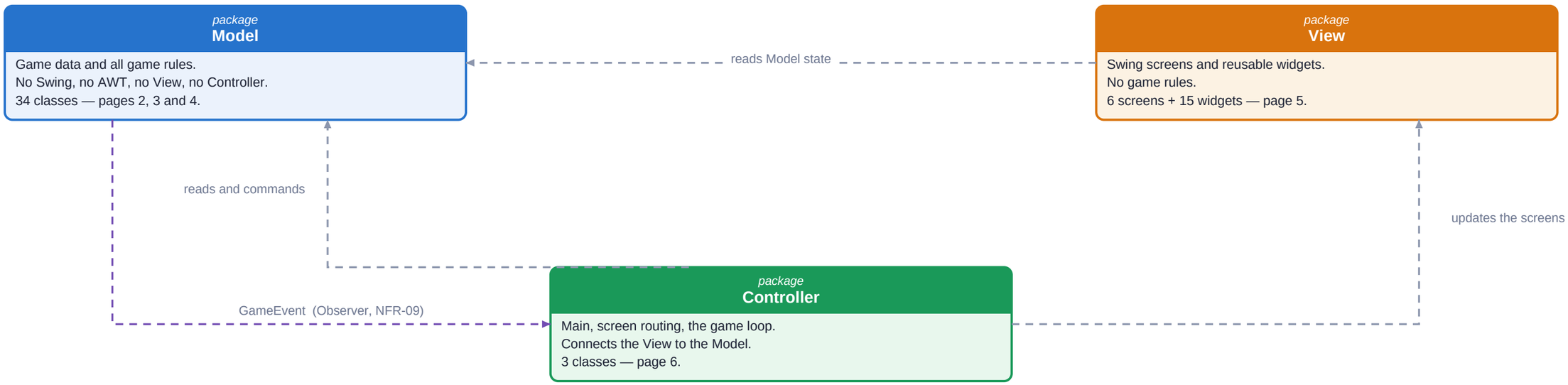
Educational Ludo — Sharks

Class diagram of the system · Iteration 1 implemented, Iterations 2–3 planned

Owner: Mohamed Zereny · Reviewer: Rakan Ziyan · SRS: NFR-08 (MVC), NFR-09 (design patterns), NFR-20 (packages)

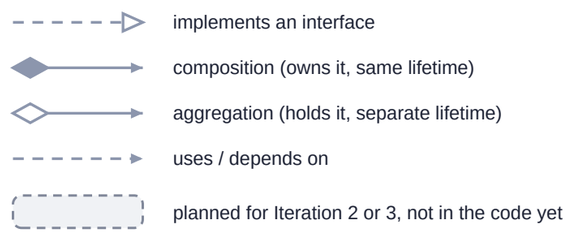
Pages 2-7: structure and main relationships (editable source: Iteration1/docs/CLASS_DIAGRAM.md). Pages 8-15: every class with all its fields and methods, and every relationship, generated from the code.



The dependency rule (NFR-08, NFR-20) is enforced by a test, not only documented

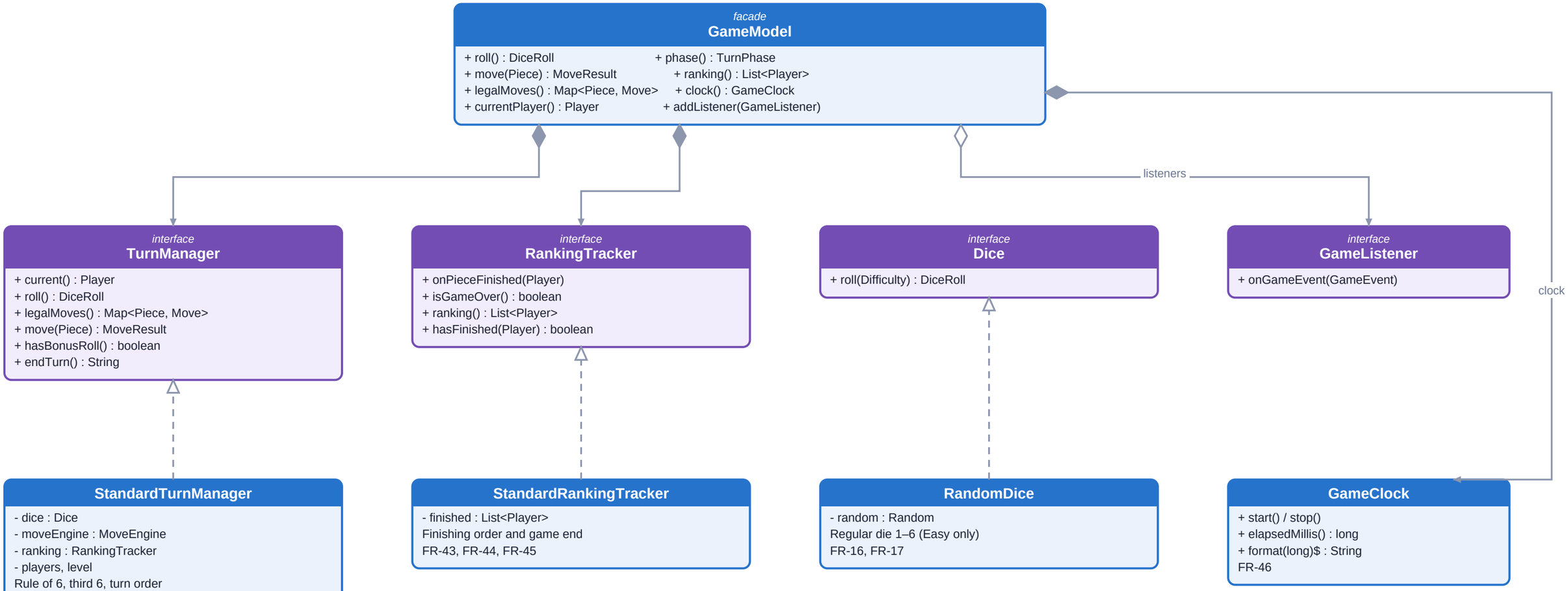
- Model may not import View or Controller, and may not import Swing or AWT.
- View may not import Controller.
- Only Model, View and Controller exist as top-level packages.
- Checked by Iteration1/test/Architecture/MvcDependencyTest.java — the build fails if it is broken.
- The Model never names the View or the Controller. It publishes a GameEvent to whoever registered as a GameListener,
- and the Controller passes the change on to the screens. That is why no arrow above leaves the Model.

Notation



Model — game flow

The GameModel facade and the contracts frozen at checkpoint CP1, each with one Standard implementation



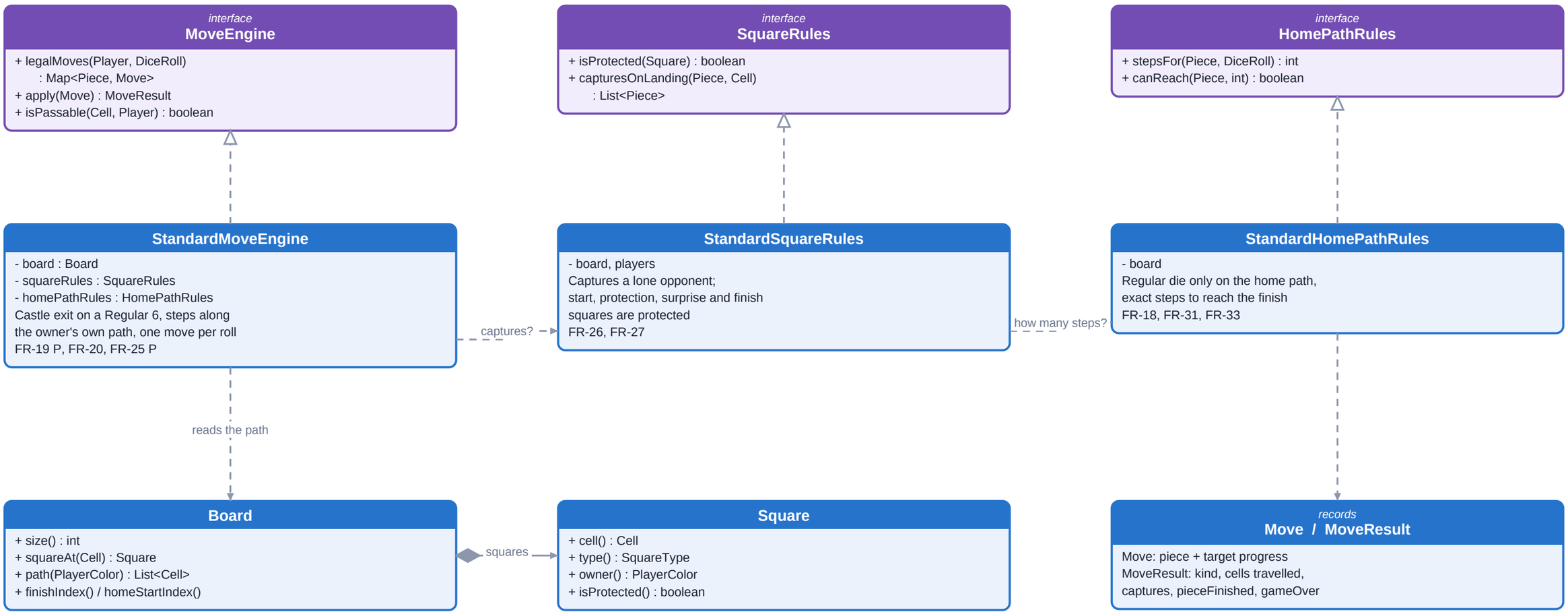
Why interfaces: the six contracts above were frozen at CP1 so the three developers could work in parallel

- Changing one of them needs a change request agreed by all three members (CONTRIBUTING.md).
- MoveEngine, SquareRules and HomePathRules are the other three contracts — page 3.
- GameSettings, Player and Piece, the data a game is built from — page 4.
- StandardTurnManager holds Dice, MoveEngine and RankingTracker as fields (listed in its box above).

Plain data types, drawn where they are used: Cell, DiceRoll, Move, MoveResult, GameEvent, PlayerColor, SquareType, DieType, Direction, Difficulty, TurnPhase.

Model — movement rules

How one legal move is worked out: the engine asks two rule objects and never decides a square rule itself



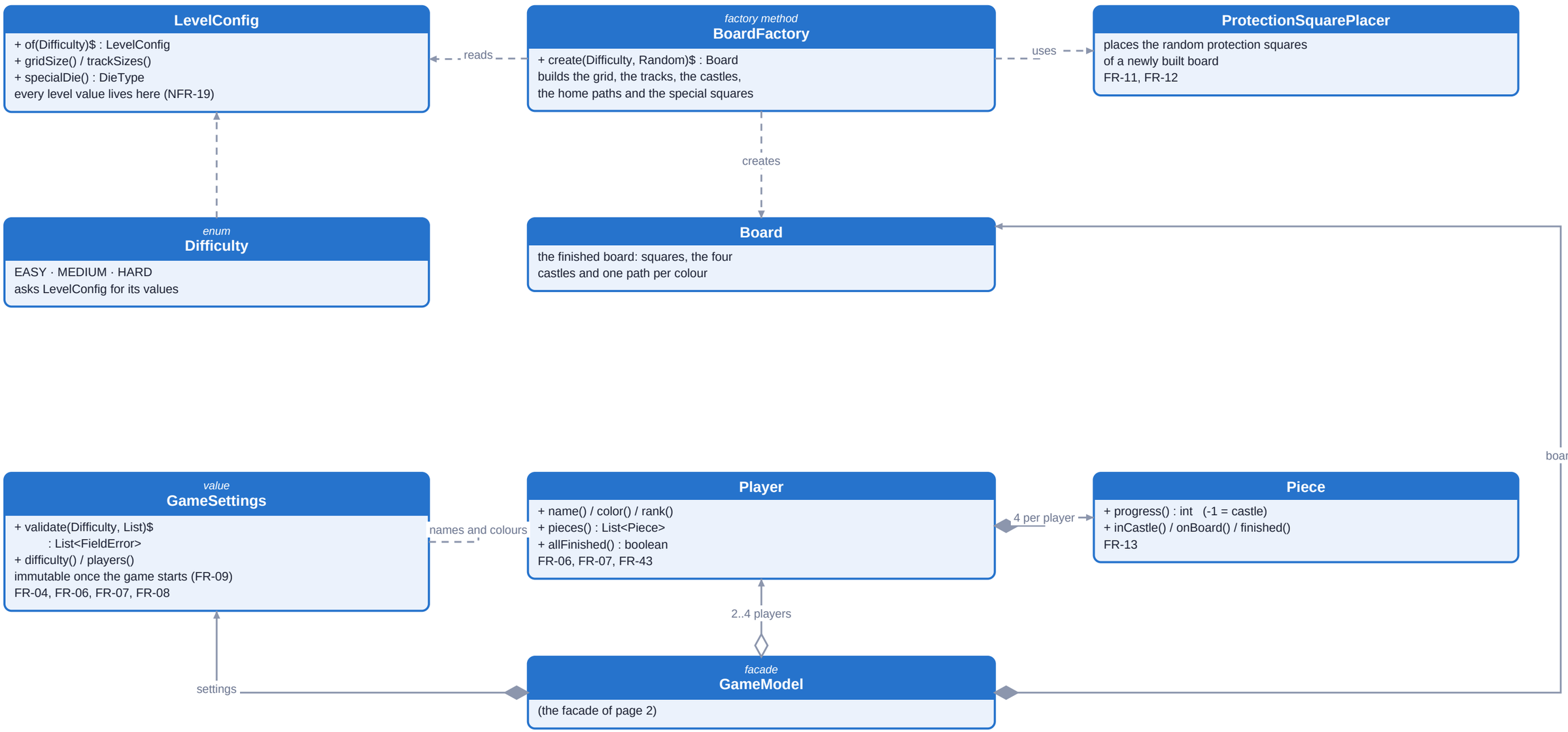
One question, three owners — this split is what let three people work on movement at the same time

- StandardMoveEngine (Task 16, Rakan) decides whether a piece may move and applies the move.
- StandardSquareRules (Task 17, Shafeek) decides capture and protection.
- StandardHomePathRules (Task 18, Shafeek) decides the steps and the exact finish.
- Double Blocks (FR-28..FR-30) arrive in Iteration 2; isPassable() is already the hook for them.

The engine holds the two rule objects behind their interfaces, so either can be replaced without touching the engine.

Model — building a board, and the data of a game

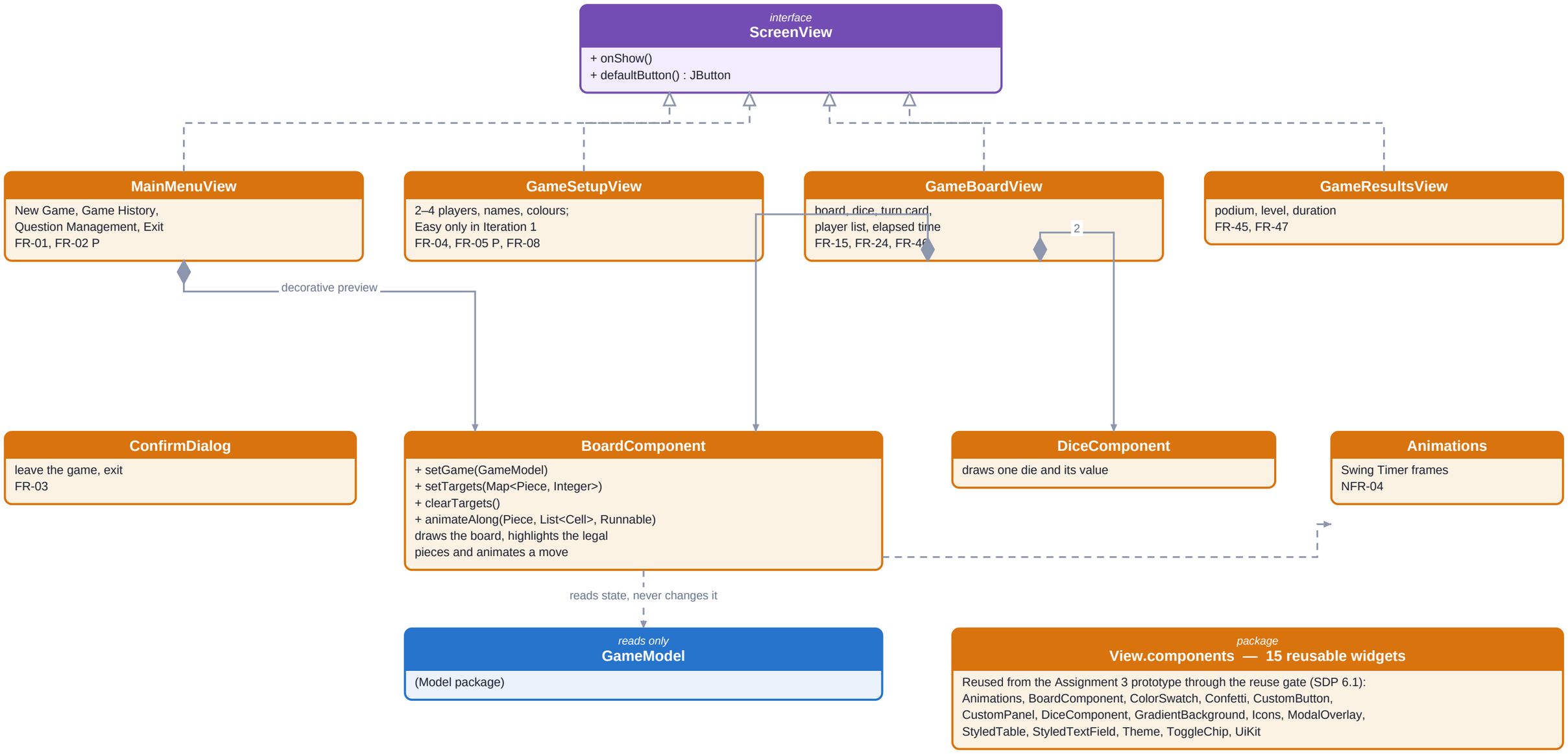
Factory Method (NFR-09) and the settings, players and pieces a game is created from



Factory Method: callers ask BoardFactory for a board of a level and never build one themselves, so Medium and Hard need no change at the call sites (NFR-09, NFR-19).

View — screens and widgets

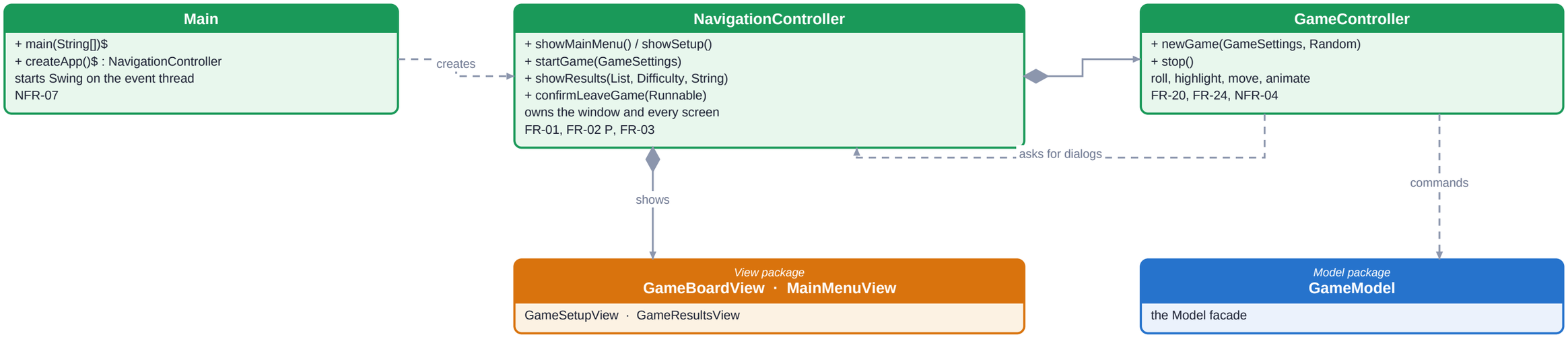
Four screens behind one interface; the board widget draws the Model and never decides a game rule



The View imports Model but never Controller. The Controller registers the listeners that refresh these screens, so the Model stays free of any screen.

Controller — entry point, routing and the game loop

The only package that knows both the screens and the Model



One turn, end to end — the Observer pattern at work (NFR-09)

1. The player presses Roll on GameBoardView.
2. GameController calls GameModel.roll().
3. The Model works the turn out and publishes GameEvent ROLLED — and TURN_PASSED plus TURN_CHANGED when nothing can move (FR-23).
4. GameController, registered as a GameListener, hears the event and asks the Model for the legal moves.
5. It hands them to BoardComponent.setTargets(...), which highlights exactly those pieces (FR-24).
6. The player clicks a highlighted piece. A piece that is not highlighted cannot be clicked at all.
7. GameController calls GameModel.move(piece), animates the result square by square (NFR-04) and clears the highlight.
8. At the end the Model publishes PLAYER_FINISHED and GAME_OVER, and the Controller shows GameResultsView with the ranking and the duration (FR-45, FR-46, FR-47).

Step 3 is the reason no arrow points from the Model to the View or the Controller: the Model publishes an event, it never calls a screen.

Design patterns, and what comes next

NFR-09 · Factory Method, Observer, State and Singleton · the classes planned for Iterations 2 and 3

Factory Method `BoardFactory.create(Difficulty, Random)`	Builds the whole board for a level: grid, tracks, castles, home paths and the random protection squares. All level data comes from LevelConfig.	**Implemented — Iteration 1**
Observer `GameModel + GameListener + GameEvent`	GameModel publishes STATE_CHANGED, ROLLED, MOVED, TURN_PASSED, TURN_CHANGED, PLAYER_FINISHED and GAME_OVER. This is what keeps the Model free of any View import.	**Implemented — Iteration 1**
State `TurnPhase + StandardTurnManager`	AWAITING_ROLL → AWAITING_MOVE → MOVING or AWAITING_ROLL → GAME_OVER. Today an enum, with the transitions written inside the turn manager.	**Partly — becomes State classes in Iteration 2**
Singleton `GameHistory (planned)`	One shared access point to the game-history CSV, so the file cannot be opened twice at the same time.	**Planned — Iteration 3**

Planned for Iterations 2 and 3 — and where each new class attaches

planned · I2
TurnState
+ onRoll() / onMove()
replaces the TurnPhase enum
attaches to StandardTurnManager

planned · I2
QuestionBank / Question
+ nextQuestion(Difficulty)
FR-34..FR-38
attaches to StandardMoveEngine

planned · I2
SurpriseRules
+ outcomeFor(Piece)
FR-39..FR-42
attaches to StandardMoveEngine

planned · I3 · singleton
GameHistory / GameRecord
+ instance()$: GameHistory
+ save(GameRecord) / all()
attaches to GameModel

Iteration 2 also adds: Double Block rules inside StandardMoveEngine, Boost and UltraBoost in RandomDice, the Medium and Hard boards, and SurpriseOutcome.
Iteration 3 also adds: CSV storage for finished games, AdminAuthenticator and the question-management screens.

None of the four classes on this row exists in the code yet. They are drawn dashed, to show only where they will attach.

Complete class reference — Model (1/3)

Every class of package Model with all its fields and methods, generated from the compiled code (source as of commit 18b21c5)

Notation: + public - private # protected ~ package · underlined = static · italic = abstract · first compartment: fields (or enum literals), second: constructors and methods · purple = interface

Board

- difficulty : Difficulty
- size : int
- offset : int
- squares : Map<Cell, Square>
- paths : Map<PlayerColor, List<Cell>>
- corners : Map<PlayerColor, Board.Corner>

~ Board(Difficulty, int, int, Map<Cell, Square>, Map<PlayerColor, List<Cell>>, Map<PlayerColor, Board.Corner>)
+ difficulty() : Difficulty
+ size() : int
+ offset() : int
+ squares() : Collection<Square>
+ squareAt(Cell) : Square
+ path(PlayerColor) : List<Cell>
+ finishIndex() : int
+ homeStartIndex() : int
+ cellAt(PlayerColor, int) : Cell
+ startCell(PlayerColor) : Cell
+ corner(PlayerColor) : Board.Corner
+ colorAt(Board.Corner) : PlayerColor
+ count(SquareType) : long
+ countQuestions(Difficulty) : long
+ trackSquareCount() : int

GameClock

- MILLIS_PER_SECOND : long
- SECONDS_PER_MINUTE : long
- SECONDS_PER_HOUR : long
- timeSource : LongSupplier
- startedAt : long
- frozenMillis : long
- highWaterMillis : long
- running : boolean

+ GameClock(LongSupplier)
+ start() : void
+ stop() : void
+ running() : boolean
+ elapsedMillis() : long
+ format(long) : String
~ timeSource() : LongSupplier

«interface»
GameModel.EngineFactory

+ create(Board, List<Player>, SquareRules, HomePathRules) : MoveEngine

«record»
GameSettings.PlayerSetup

- name : String
- color : PlayerColor

+ name() : String
+ color() : PlayerColor

«enumeration»
Board.Corner

TOP_LEFT, TOP_RIGHT, BOTTOM_RIGHT, BOTTOM_LEFT

«interface»
Dice

+ roll(Difficulty) : DiceRoll

«enumeration»
Difficulty

EASY, MEDIUM, HARD

- label : String

- Difficulty(String)
+ label() : String
+ config() : LevelConfig
+ gridSize() : int
+ outerRings() : int
+ trackSizes() : int[]
+ totalTrackSquares() : int
+ trackFormula() : String
+ protectionSquares() : int
+ arrowSquares() : int
+ questionSquares() : int
+ questionsPerLevel() : int
+ surpriseSquares() : int
+ specialDie() : DieType
+ questionSteps() : int
+ diceDescription() : String

«interface»
GameListener

+ onGameEvent(GameEvent) : void

GameSettings

+ MIN_PLAYERS : int
+ MAX_PLAYERS : int
+ MAX_NAME_LENGTH : int
+ FIELD_DIFFICULTY : String
+ FIELD_PLAYERS : String
+ FIELD_NAME : String
+ FIELD_COLOR : String
- difficulty : Difficulty
- players : List<GameSettings.PlayerSetup>

+ GameSettings(Difficulty, List<GameSettings.PlayerSetup>)
+ difficulty() : Difficulty
+ players() : List<GameSettings.PlayerSetup>
+ playerCount() : int
+ validate(Difficulty, List<GameSettings.PlayerSetup>) : List<GameSettings.FieldError>
~ cleanName(String) : String
- isInvisible(char) : boolean
- sameName(String, String) : boolean

BoardFactory

- BoardFactory()
+ create(Difficulty, Random) : Board
~ cornersFor(Difficulty) : Map<PlayerColor, Board.Corner>
~ topLeftPath(int, int) : List<Cell>
~ ring(int, int) : List<Cell>
~ innerTrack(int) : List<Cell>
- line(List<Cell>, int, int, int, int) : void
- placeSpecialSquares(LevelConfig, Map<Cell, Square>, Random) : void

«enumeration»
DieType

NONE, REGULAR, BOOST, ULTRA_BOOST

- label : String
- faces : int

- DieType(String, int)
+ label() : String
+ faces() : int

«enumeration»
Direction

NORTH, EAST, SOUTH, WEST

- Direction()
+ rotateClockwise(int) : Direction

«enumeration»
GameEvent.Type

STATE_CHANGED, ROLLED, MOVED, TURN_PASSED, TURN_CHANGED, PLAYER_FINISHED, GAME_OVER

GameModel

- settings : GameSettings
- board : Board
- players : List<Player>
- turns : TurnManager
- ranking : RankingTracker
- clock : GameClock
- listeners : List<GameListener>

+ GameModel(GameSettings, Random)
+ GameModel(GameSettings, Random, Dice, LongSupplier)
~ GameModel(GameSettings, Random, Dice, LongSupplier, GameModel.EngineFactory)
+ addListener(GameListener) : void
+ removeListener(GameListener) : void
- fire(GameEvent) : void
+ roll() : DiceRoll
+ move(Piece) : MoveResult
- afterTurnEnded() : void
+ settings() : GameSettings
+ difficulty() : Difficulty
+ board() : Board
+ players() : List<Player>
+ playerOf(PlayerColor) : Player
+ currentPlayer() : Player
+ phase() : TurnPhase
+ lastRoll() : DiceRoll
+ legalMoves() : Map<Piece, Move>
+ hasBonusRoll() : boolean
+ isGameOver() : boolean
+ ranking() : List<Player>
+ clock() : GameClock
+ refresh() : void

«record»
Cell

- row : int
- col : int

+ Cell(int, int)
+ rotate(int, int) : Cell
+ row() : int
+ col() : int

«record»
DiceRoll

- regular : int
- specialType : DieType
- special : int
- movement : int
- surpriseRoll : boolean
- cancelled : boolean

+ DiceRoll(int, DieType, int, int, boolean, boolean)
+ regularOnly(int) : DiceRoll
+ hasSpecial() : boolean
+ regular() : int
+ specialType() : DieType
+ special() : int
+ movement() : int
+ surpriseRoll() : boolean
+ cancelled() : boolean

«record»
GameEvent

- type : GameEvent.Type
- player : Player
- roll : DiceRoll
- move : MoveResult
- message : String

+ GameEvent(GameEvent.Type, Player, DiceRoll, MoveResult, String)
+ stateChanged(Player) : GameEvent
+ type() : GameEvent.Type
+ player() : Player
+ roll() : DiceRoll
+ move() : MoveResult
+ message() : String

«record»
GameSettings.FieldError

- player : int
- field : String
- message : String

+ player() : int
+ field() : String
+ message() : String

«interface»
HomePathRules

+ stepsFor(Piece, DiceRoll) : int
+ canReach(Piece, int) : boolean

Complete class reference — Model (2/3)

Every class of package Model with all its fields and methods, generated from the compiled code (source as of commit 18b21c5)

Notation: + public - private # protected ~ package · underlined = static · italic = abstract · first compartment: fields (or enum literals), second: constructors and methods · purple = interface

LevelConfig

- HOME_PATH_SQUARES : int
- STARTING_SQUARES : int
- CLASSIC_GRID_SIZE : int
- LEVELS : Map<Difficulty, LevelConfig>
- gridSize : int
- trackSizes : int[]
- protectionSquares : int
- arrowSquares : int
- questionSquares : int
- surpriseSquares : int
- specialDie : DieType
- questionSteps : int

- LevelConfig(int, int[], int, int, int, int, DieType, int)
+ of(Difficulty) : LevelConfig
+ gridSize() : int
+ outerRings() : int
+ trackSizes() : int[]
+ totalTrackSquares() : int
+ trackFormula() : String
+ homePathSquares() : int
+ startingSquares() : int
+ protectionSquares() : int
+ arrowSquares() : int
+ questionSquares() : int
+ questionsPerLevel() : int
+ surpriseSquares() : int
+ specialDie() : DieType
+ questionSteps() : int
+ diceDescription() : String

ProtectionSquarePlacer

- ProtectionSquarePlacer()
+ place(Difficulty, Collection<Square>, Random) : List<Square>
+ place(int, Collection<Square>, Random) : List<Square>
+ isFree(Square) : boolean

«interface»
SquareRules

+ isProtected(Square) : boolean
+ capturesOnLanding(Piece, Cell) : List<Piece>

StandardMoveEngine

implements MoveEngine

~ EXIT_ROLL : int
- board : Board
- players : List<Player>
- squareRules : SquareRules
- homePathRules : HomePathRules
- offered : Map<Piece, Move>

+ StandardMoveEngine(Board, List<Player>, SquareRules, HomePathRules)
+ legalMoves(Player, DiceRoll) : Map<Piece, Move>
- moveFor(Piece, Player, DiceRoll) : Move
+ apply(Move) : MoveResult
+ isPassable(Cell, Player) : boolean
- isGameOver() : boolean
- requireCapturable(Piece, Piece, Cell) : void
- requireInGame(Player) : void
- ownerOf(Piece) : Player
~ board() : Board
~ players() : List<Player>
~ squareRules() : SquareRules
~ homePathRules() : HomePathRules

«record»
Move

- piece : Piece
- targetProgress : int

+ Move(Piece, int)
+ piece() : Piece
+ targetProgress() : int

«enumeration»
MoveResult.Kind

STEPS, EXIT_CASTLE, TO_CASTLE, TO_FINISH, NONE

Piece

+ IN_CASTLE : int
- color : PlayerColor
- index : int
- finishIndex : int
- progress : int

+ Piece(PlayerColor, int, int)
+ color() : PlayerColor
+ index() : int
+ progress() : int
+ finishIndex() : int
~ setProgress(int) : void
+ inCastle() : boolean
+ finished() : boolean
+ onBoard() : boolean
+ toString() : String

RandomDice

implements Dice

- random : Random

+ RandomDice(Random)
+ roll(Difficulty) : DiceRoll
~ random() : Random

«enumeration»
SquareType

TRACK, START, ARROW, PROTECTION, QUESTION, SURPRISE, HOME_PATH

- SquareType()
+ isProtected() : boolean

StandardRankingTracker

implements RankingTracker

- players : List<Player>
- finished : List<Player>

+ StandardRankingTracker(List<Player>)
+ onPieceFinished(Player) : void
+ isGameOver() : boolean
+ ranking() : List<Player>
+ hasFinished(Player) : boolean
- remaining() : List<Player>
~ players() : List<Player>

«interface»
MoveEngine

+ legalMoves(Player, DiceRoll) : Map<Piece, Move>
+ apply(Move) : MoveResult
+ isPassable(Cell, Player) : boolean

«record»
MoveResult.Capture

- piece : Piece
- from : Cell

+ piece() : Piece
+ from() : Cell

Player

+ PIECES_PER_PLAYER : int
- name : String
- color : PlayerColor
- pieces : List<Piece>
- skipNextTurn : boolean
- rank : int

+ Player(String, PlayerColor, int)
+ name() : String
+ color() : PlayerColor
+ pieces() : List<Piece>
+ finishedCount() : int
+ allFinished() : boolean
+ skipNextTurn() : boolean
~ setSkipNextTurn(boolean) : void
+ rank() : int
~ setRank(int) : void
+ toString() : String

Square

- cell : Cell
- ring : int
- type : SquareType
- owner : PlayerColor
- arrowDirection : Direction
- questionLevel : Difficulty
- tinted : boolean

~ Square(Cell, int)
+ cell() : Cell
+ ring() : int
+ type() : SquareType
+ owner() : PlayerColor
+ arrowDirection() : Direction
+ questionLevel() : Difficulty
+ isTinted() : boolean
+ isProtected() : boolean
~ setType(SquareType) : void
~ setOwner(PlayerColor) : void
~ setArrowDirection(Direction) : void
~ setQuestionLevel(Difficulty) : void
~ setTinted(boolean) : void

«record»
MoveResult

- piece : Piece
- kind : MoveResult.Kind
- fromProgress : int
- toProgress : int
- cells : List<Cell>
- fromCell : Cell
- captures : List<MoveResult.Capture>
- landedOn : Square
- pieceFinished : boolean
- playerFinished : boolean
- gameOver : boolean

+ MoveResult(Piece, MoveResult.Kind, int, int, List<Cell>, Cell, List<MoveResult.Capture>, Square, boolean, boolean, boolean)
~ none(Piece) : MoveResult
+ piece() : Piece
+ kind() : MoveResult.Kind
+ fromProgress() : int
+ toProgress() : int
+ cells() : List<Cell>
+ fromCell() : Cell
+ captures() : List<MoveResult.Capture>
+ landedOn() : Square
+ pieceFinished() : boolean
+ playerFinished() : boolean
+ gameOver() : boolean

«enumeration»
PlayerColor

RED, GREEN, YELLOW, BLUE

- label : String

- PlayerColor(String)
+ label() : String

«interface»
RankingTracker

+ onPieceFinished(Player) : void
+ isGameOver() : boolean
+ ranking() : List<Player>
+ hasFinished(Player) : boolean

StandardHomePathRules

implements HomePathRules

- board : Board

+ StandardHomePathRules(Board)
+ stepsFor(Piece, DiceRoll) : int
+ canReach(Piece, int) : boolean
~ board() : Board

StandardSquareRules

implements SquareRules

- board : Board
- players : List<Player>

+ StandardSquareRules(Board, List<Player>)
+ isProtected(Square) : boolean
+ capturesOnLanding(Piece, Cell) : List<Piece>
~ board() : Board
~ players() : List<Player>

Complete class reference — Model (3/3)

Every class of package Model with all its fields and methods, generated from the compiled code (source as of commit 18b21c5)

Notation: + public - private # protected ~ package · underlined = static · italic = abstract · first compartment: fields (or enum literals), second: constructors and methods · purple = interface

StandardTurnManager

implements TurnManager

~ SIXES_LIMIT : int
- players : List<Player>
- dice : Dice
- moveEngine : MoveEngine
- ranking : RankingTracker
- level : Difficulty
- currentIndex : int
- phase : TurnPhase
- lastRoll : DiceRoll
- sixesInRow : int
- bonusRoll : boolean
- cachedMoves : Map<Piece, Move>

+ StandardTurnManager(List<Player>, Difficulty, Dice, MoveEngine, RankingTracker)
+ current() : Player
+ phase() : TurnPhase
+ lastRoll() : DiceRoll
+ roll() : DiceRoll
+ canRoll() : boolean
+ legalMoves() : Map<Piece, Move>
+ move(Piece) : MoveResult
+ hasBonusRoll() : boolean
+ endTurn() : String
~ dice() : Dice
~ moveEngine() : MoveEngine
~ ranking() : RankingTracker
~ level() : Difficulty

«interface»
TurnManager

+ current() : Player
+ phase() : TurnPhase
+ lastRoll() : DiceRoll
+ roll() : DiceRoll
+ legalMoves() : Map<Piece, Move>
+ move(Piece) : MoveResult
+ hasBonusRoll() : boolean
+ endTurn() : String

«enumeration»
TurnPhase

AWAITING_ROLL, ROLLING, AWAITING_MOVE, MOVING, RESOLVING, GAME_OVER

- TurnPhase()

Complete class reference — View — screens

Every class of package View with all its fields and methods, generated from the compiled code (source as of commit 18b21c5)

Notation: + public - private # protected ~ package · underlined = static · italic = abstract · first compartment: fields (or enum literals), second: constructors and methods · purple = interface

ConfirmDialog

extends CustomPanel

- confirm : CustomButton
- cancel : CustomButton

+ ConfirmDialog(String, String, String, String, boolean, Icons.Type, Runnable, Runnable)
+ getPreferredSize() : Dimension
+ confirmButton() : CustomButton
+ cancelButton() : CustomButton

GameBoardView.TurnCard

extends JPanel

- eyebrowLabel : JLabel
- name : JLabel
- detail : JLabel
- pawn : JLabel
- color : PlayerColor

~ show(Player, int, int, boolean) : void
paintComponent(Graphics) : void

GameSetupView

extends GradientBackground
implements ScreenView

- DEFAULT_COLORS : PlayerColor[]
- back : CustomButton
- start : CustomButton
- countChips : Map<Integer, ToggleChip>
- difficultyCards : Map<Difficulty, GameSetupView.DifficultyCard>
- rows : List<GameSetupView.PlayerRow>
- summary : JLabel
- playerCount : int
- difficulty : Difficulty
- startHandler : BiFunction<Difficulty, List<GameSettings.PlayerSetup>, List<GameSettings.FieldError>>

+ GameSetupView()
- optionsCard() : JComponent
- playersCard() : JComponent
- section(String) : JLabel
+ setStartHandler(BiFunction<Difficulty, List<GameSettings.PlayerSetup>, List<GameSettings.FieldError>>) : void
+ setBackAction(Runnable) : void
+ setPlayerCount(int) : void
+ playable(Difficulty) : boolean
+ setDifficulty(Difficulty) : void
+ difficulty() : Difficulty
+ setPlayerName(int, String) : void
+ setPlayerColor(int, PlayerColor) : void
+ playerCount() : int
+ visibleErrors() : List<String>
- firstFreeColor(int) : PlayerColor
- refreshSwatches() : void
- submit() : void
+ defaultButton() : JButton
+ onShow() : void

MainMenuView.Option

extends JPanel

- RIGHT_MARGIN : int
- button : JButton
- tag : MainMenuView.ComingSoonTag

+ doLayout() : void
+ isOptimizedDrawingEnabled() : boolean
+ getPreferredSize() : Dimension
+ getMinimumSize() : Dimension
+ getMaximumSize() : Dimension

ConfirmDialog.IconBadge

extends JComponent

- icon : Icons.Type
- color : Color

paintComponent(Graphics) : void

GameBoardView.PlayerRow

extends JPanel

- active : boolean
- color : PlayerColor

paintComponent(Graphics) : void

GameResultsView

extends JPanel
implements ScreenView

- confetti : Confetti
- podium : GameResultsView.Podium
- chips : JPanel
- extraPlaces : JPanel
- saved : JLabel
- playAgain : CustomButton
- menu : CustomButton

+ GameResultsView()
- center(JComponent) : JComponent
+ isOptimizedDrawingEnabled() : boolean
+ setActions(Runnable, Runnable) : void
+ showResults(List<Player>, Difficulty, String) : void
+ onShow() : void
+ defaultButton() : JButton

GameSetupView.PlayerRow.Badge

extends JComponent

- number : int

paintComponent(Graphics) : void

MainMenuView

extends GradientBackground
implements ScreenView

- COLUMN_GAP : int
- SLACK : int
- newGame : CustomButton
- history : CustomButton
- questions : CustomButton
- exit : CustomButton
- hero : MainMenuView.Hero
- column : JComponent

+ MainMenuView(GameModel)
+ doLayout() : void
- menuColumn() : JComponent
- gap(int) : Component
- left(JComponent) : JComponent
- menuButton(String, Icons.Type, CustomButton.Variant, String) : CustomButton
+ setActions(Runnable, Runnable, Runnable, Runnable) : void
+ defaultButton() : JButton

«interface»
ScreenView

+ onShow() : void
+ defaultButton() : JButton

GameBoardView

extends GradientBackground
implements ScreenView

- LEFT_WIDTH : int
- RIGHT_WIDTH : int
- board : BoardComponent
- regularDie : DiceComponent
- specialDie : DiceComponent
- times : JLabel
- regularCaption : JLabel
- specialCaption : JLabel
- specialBox : JPanel
- formula : JLabel
- formulaDetail : JLabel
- roll : CustomButton
- back : CustomButton
- timer : JLabel
- levelPill : UiKit.Pill
- boardSize : JLabel
- trackInfo : JLabel
- playerList : JPanel
- legend : JPanel
- turnCard : GameBoardView.TurnCard
- statusIcon : JLabel
- statusText : JLabel
- state : GameModel
- rollAction : Runnable

+ GameBoardView()
- caption(String) : JLabel
- eyebrow() : Font
- leftPanel() : JComponent
- rightPanel() : JComponent
- glass(LayoutManager) : CustomPanel
+ setActions(Runnable, Runnable) : void
+ bind(GameModel) : void
- specialLabel(DieType) : String
- buildLegend() : void
- legendRow(Icon, String, int) : JComponent
+ refresh() : void
+ resetDice() : void
+ showRoll(DiceRoll) : void
+ setStatus(String, GameBoardView.StatusKind) : void
+ statusText() : String
+ setElapsed(String) : void
+ setControls(boolean, boolean) : void
+ board() : BoardComponent
+ regularDie() : DiceComponent
+ specialDie() : DiceComponent
+ rollButton() : CustomButton
+ onShow() : void

MainMenuView.Hero

extends JPanel

- DESIGN_WIDTH : int
- DESIGN_HEIGHT : int
- MIN_WIDTH : int
- MIN_HEIGHT : int
- DESIGN_BOARD : int
- board : BoardComponent
- die : DiceComponent
- boost : DiceComponent
- budgetWidth : int
- budgetHeight : int

~ setBudget(int, int) : void
+ getPreferredSize() : Dimension
+ getMinimumSize() : Dimension
+ getMaximumSize() : Dimension
- clamp(int, int, int) : int
+ doLayout() : void
- scaled(int, double) : int
+ isOptimizedDrawingEnabled() : boolean
paintComponent(Graphics) : void

GameBoardView.LegendIcon

implements Icon

- type : SquareType

+ paintIcon(Component, Graphics, int, int) : void
+ getIconWidth() : int
+ getIconHeight() : int

«enumeration»
GameBoardView.StatusKind

INFO, SUCCESS, WARNING, ERROR

GameResultsView.Podium

extends JComponent

- ranking : List<Player>

~ setRanking(List<Player>) : void
paintComponent(Graphics) : void
- placeLabel(int) : String

GameSetupView.DifficultyCard

extends JToggleButton

- level : Difficulty
- hover : boolean

- line1() : String
- line2() : String
paintComponent(Graphics) : void

GameSetupView.PlayerRow

extends JPanel

- name : StyledTextField
- counter : JLabel
- error : JLabel
- swatches : List<ColorSwatch>
- badge : GameSetupView.PlayerRow.Badge
- color : PlayerColor

~ setColor(PlayerColor) : void
- changed() : void
~ showError(String, String) : void
~ clearError() : void

MainMenuView.ComingSoonTag

extends JComponent

- TEXT : String
- FONT_SIZE : float
- PADDING_X : int
- PADDING_Y : int

+ getPreferredSize() : Dimension
+ contains(int, int) : boolean
paintComponent(Graphics) : void

MainMenuView.Title

extends JComponent

- DESIGN_WIDTH : int
- DESIGN_HEIGHT : int
- TOP_LINE : String
- BOTTOM_LINE : String
- TOP_SHADOW_X : int
- TOP_SHADOW_Y : int
- BOTTOM_SHADOW_X : int
- BOTTOM_SHADOW_Y : int
- LETTER_GAP : int
- top : Font
- bottom : Font
- letterX : int[]
- baselineTop : int
- baselineBottom : int

paintComponent(Graphics) : void

Complete class reference — View.components — widgets (1/2)

Every class of package View.components with all its fields and methods, generated from the compiled code (source as of commit 18b21c5)

Notation: + public - private # protected ~ package · underlined = static · italic = abstract · first compartment: fields (or enum literals), second: constructors and methods · purple = interface

Animations

- + instant : boolean

- - Animations()
- + animate(int, DoubleConsumer, Runnable) : Timer
- + later(int, Runnable) : void
- + easeOut(double) : double
- + easeInOut(double) : double

CustomButton

extends JButton

- - variant : CustomButton.Variant
- - accent : Color
- - hover : float
- - hoverTarget : float
- - hoverTimer : Timer
- - radius : int

- + CustomButton(String, CustomButton.Variant)
- + CustomButton(String, Icon, CustomButton.Variant)
- + setAccent(Color) : void
- + setRadius(int) : void
- + getPreferredSize() : Dimension
- - fillColor() : Color
- - textColor() : Color
- - isLight(Color) : boolean
- # paintComponent(Graphics) : void

«enumeration»
CustomPanel.Style

CARD, GLASS, DASHED, SOLID

GradientBackground

extends JPanel

- + GradientBackground(LayoutManager)
- # paintComponent(Graphics) : void
- - glow(Graphics2D, double, double, double, Color) : void

Icons

- - Icons()
- + get(Icons.Type, int, Color) : Icon
- + paint(Graphics2D, Icons.Type, double, double, double, Color) : void
- + star(double, double, double, double, int) : Shape
- + gift(Graphics2D, double, double, double, Color) : void
- + glyph(String, Font, double, double, double) : Shape
- - dot(Graphics2D, double, double, double) : void
- - poly(double...) : Shape
- - poly(double, double, double, double, double, double, boolean) : Shape
- - poly(double, double, double, double, double, double, double, double, boolean) : Shape
- - poly(boolean, double...) : Shape

StyledTable

- - STRIPE : Color
- - SELECTION : Color

- - StyledTable()
- + style(JTable) : void
- + scroll(JTable) : JScrollPane

StyledTable.TextRenderer

extends DefaultTableCellRenderer

- + getTableCellRendererComponent(JTable, Object, boolean, boolean, int, int) : Component

BoardComponent

extends JComponent

- - SPOTS : int[][]
- - state : GameModel
- - interactive : boolean
- - showActiveGlow : boolean
- - listener : BoardComponent.PieceListener
- - targets : Map<Piece, Integer>
- - hovered : Piece
- - overrides : Map<Piece, Point2D.Double>
- - cache : BufferedImage
- - cacheW : int
- - cacheH : int
- - cacheScale : double
- - pulse : Timer
- - pulsePhase : double
- - bannerText : String
- - bannerColor : Color
- - bannerStart : long
- - n : int
- - cell : double
- - ox : double
- - oy : double

- + BoardComponent()
- + setGame(GameModel) : void
- + setInteractive(boolean) : void
- + setPieceListener(BoardComponent.PieceListener) : void
- + setTargets(Map<Piece, Integer>) : void
- + clearTargets() : void
- + targets() : Map<Piece, Integer>
- + setHovered(Piece) : void
- + refreshStatic() : void
- + showBanner(String, Color) : void
- + addNotify() : void
- + removeNotify() : void
- - updatePulse() : void
- + cellCenter(Cell) : Point2D.Double
- - castleOrigin(PlayerColor) : int[]
- + castleSpot(Piece) : Point2D.Double
- + finishSpot(Piece) : Point2D.Double
- + restingPosition(Piece) : Point2D.Double
- + holdPiece(Piece, Point2D.Double) : void
- + animateAlong(Piece, List<Cell>, Runnable) : void
- + animateJump(Piece, Point2D.Double, Point2D.Double, Runnable) : void
- - layoutBoard() : void
- - px(double) : double
- - py(double) : double
- # paintComponent(Graphics) : void
- - paintStatic(Graphics2D) : void
- - drawSquare(Graphics2D, Square, BasicStroke) : void
- - drawArrow(Graphics2D, double, double, Direction, Color) : void
- - castleRect(PlayerColor) : Rectangle2D
- - drawCastle(Graphics2D, PlayerColor) : void
- - drawCenter(Graphics2D) : void
- - triangle(Graphics2D, PlayerColor, double, double, double, double, double, double) : void
- - paintDynamic(Graphics2D) : void
- - drawPreview(Graphics2D, Piece, int) : void
- - drawPieces(Graphics2D) : void
- - order(Piece) : int
- - groupOffset(int, int) : double[]
- + pawnIcon(PlayerColor, int) : Icon
- + paintPawn(Graphics2D, double, double, double, PlayerColor) : void
- - drawBanner(Graphics2D) : void
- + pieceAt(Point) : Piece
- + screenPosition(Piece) : Point

StyledTable.HeaderRenderer

extends DefaultTableCellRenderer

- + getTableCellRendererComponent(JTable, Object, boolean, boolean, int, int) : Component

«interface»
BoardComponent.PieceListener

- + pieceClicked(Piece) : void

Confetti

extends JComponent

- - random : Random
- - particles : List<Confetti.Particle>
- - timer : Timer
- - start : long

- + Confetti()
- + contains(int, int) : boolean
- + start() : void
- + stop() : void
- - step() : void
- # paintComponent(Graphics) : void

«enumeration»
CustomButton.Variant

PRIMARY, SECONDARY, LIGHT, GHOST, DANGER, DEMO

CustomPanel

extends JPanel

- - SHADOW : int
- - style : CustomPanel.Style
- - fill : Color
- - outline : Color
- - radius : int

- + CustomPanel(LayoutManager, CustomPanel.Style)
- + setPadding(int) : void
- + setPadding(int, int, int, int) : void
- + setFill(Color) : void
- + setOutline(Color) : void
- + setRadius(int) : void
- # paintComponent(Graphics) : void

«enumeration»
Icons.Type

PLAY, HISTORY, LOCK, EXIT, BACK, ARROW_RIGHT, DICE, USER, PLUS, EDIT, TRASH, CHECK, CROSS, STAR, GIFT, QUESTION, TROPHY, CLOCK, FLAG, CASTLE, SKIP, ROCKET, INFO, WARNING, PUSH, STEPS_BACK, CROWN, BOARD, EYE, LIST, REFRESH, SHIELD

ModalOverlay

extends JPanel

- - OPEN : List<ModalOverlay>
- - host : JLayeredPane
- - card : JComponent
- - resizer : ComponentAdapter
- - alpha : float
- - onEscape : Runnable
- - onClose : Runnable

- - ModalOverlay(JLayeredPane, JComponent)
- + show(JLayeredPane, JComponent) : ModalOverlay
- + doLayout() : void
- + setOnEscape(Runnable) : void
- + setOnClose(Runnable) : void
- + card() : JComponent
- + close() : void
- + openCount() : int
- + top() : ModalOverlay
- + closeAll() : void
- + paint(Graphics) : void

ColorSwatch

extends JToggleButton

- - color : PlayerColor

- + ColorSwatch(PlayerColor)
- + playerColor() : PlayerColor
- # paintComponent(Graphics) : void

«record»
Confetti.Particle

- - x : double
- - y : double
- - vx : double
- - vy : double
- - rot : double
- - vr : double
- - w : double
- - h : double
- - color : Color

- + x() : double
- + y() : double
- + vx() : double
- + vy() : double
- + rot() : double
- + vr() : double
- + w() : double
- + h() : double
- + color() : Color

DiceComponent

extends JComponent

- - PIPS : int[][][]
- - random : Random
- - type : DieType
- - value : int
- - tilt : double
- - bounce : double
- - dimmed : boolean

- + DiceComponent(DieType, int)
- + setType(DieType) : void
- + getType() : DieType
- + setValue(int) : void
- + getValue() : int
- + setTilt(double) : void
- + setDimmed(boolean) : void
- + roll(int, Runnable) : void
- # paintComponent(Graphics) : void

«record»
Icons.VectorIcon

implements Icon

- - type : Icons.Type
- - size : int
- - color : Color

- + paintIcon(Component, Graphics, int, int) : void
- + getIconWidth() : int
- + getIconHeight() : int
- + type() : Icons.Type
- + size() : int
- + color() : Color

StyledTable.PillRenderer

extends DefaultTableCellRenderer

- - level : Difficulty
- - selected : boolean
- - row : int

- + getTableCellRendererComponent(JTable, Object, boolean, boolean, int, int) : Component
- # paintComponent(Graphics) : void

Complete class reference — View.components — widgets (2/2)

Every class of package View.components with all its fields and methods, generated from the compiled code (source as of commit 18b21c5)

Notation: + public - private # protected ~ package · underlined = static · italic = abstract · first compartment: fields (or enum literals), second: constructors and methods · purple = interface

StyledTextField

extends JTextField

- look : StyledTextField.FieldLook

+ StyledTextField(String, Icon)
+ setMaxLength(int) : void
+ setError(boolean) : void
+ hasError() : boolean
+ getPreferredSize() : Dimension
paintComponent(Graphics) : void

StyledTextField.FieldLook

- field : JTextComponent
- placeholder : String
- icon : Icon
- error : boolean

~ setError(boolean) : void
~ limit(JTextComponent, int) : void
~ paintBackground(Graphics) : void
~ paintOverlay(Graphics) : void

StyledTextField.Password

extends JPasswordField

- look : StyledTextField.FieldLook

+ setError(boolean) : void
+ getPreferredSize() : Dimension
paintComponent(Graphics) : void

Theme

+ BG_TOP : Color
+ BG_BOTTOM : Color
+ SURFACE : Color
+ SURFACE_ALT : Color
+ BORDER : Color
+ TEXT : Color
+ TEXT_MUTED : Color
+ TEXT_ON_DARK : Color
+ TEXT_ON_DARK_MUTED : Color
+ PRIMARY : Color
+ PRIMARY_DARK : Color
+ ACCENT : Color
+ SUCCESS : Color
+ DANGER : Color
+ WARNING : Color
+ BOARD_LINE : Color
+ SURPRISE : Color
+ STAR : Color
+ GOLD : Color
+ SILVER : Color
+ BRONZE : Color
- BODY_FAMILIES : String[]
- DISPLAY_FAMILIES : String[]
- MONO_FAMILIES : String[]
- bodyFamily : String
- displayFamily : String
- monoFamily : String

- Theme()
+ player(PlayerColor) : Color
+ playerDark(PlayerColor) : Color
+ playerLight(PlayerColor) : Color
+ textOn(PlayerColor) : Color
+ level(Difficulty) : Color
+ levelLight(Difficulty) : Color
+ mix(Color, Color, double) : Color
+ alpha(Color, int) : Color
- initFonts() : void
- pick(Set<String>, String[]) : String
+ font(int, float) : Font
+ display(float) : Font
+ mono(int, float) : Font
+ antialias(Graphics2D) : void
+ installLookAndFeel() : void

UiKit

- cssPixelScale : double

- UiKit()
+ label(String, Font, Color) : JLabel
+ wrap(String, int, Font, Color) : JLabel
+ setWrapped(JLabel, String, int) : void
+ ltr(String) : String
- cssPixelScale() : double
+ escape(String) : String
+ vgap(int) : Component
+ hgap(int) : Component
+ header(JComponent, String, String, JComponent) : JComponent
+ pill(String, Color, Color) : JLabel

UiKit.Pill

extends JLabel

- background : Color

+ setPillColors(Color, Color) : void
paintComponent(Graphics) : void

ToggleChip

extends JToggleButton

- onDark : boolean
- selectedColor : Color
- hover : boolean

+ ToggleChip(String, boolean)
+ setSelectedColor(Color) : void
+ getPreferredSize() : Dimension
paintComponent(Graphics) : void
- isLight(Color) : boolean

Complete class reference — Controller

Every class of package Controller with all its fields and methods, generated from the compiled code (source as of commit 18b21c5)

Notation: + public - private # protected ~ package · underlined = static · italic = abstract · first compartment: fields (or enum literals), second: constructors and methods · purple = interface

GameController

~ PASS_DELAY_MS : int
~ GAME_OVER_DELAY_MS : int
- nav : NavigationController
- model : GameModel
- view : GameBoardView
- clockTimer : Timer
- busy : boolean
- passMessage : String
- generation : int

+ GameController(NavigationController)
+ newGame(GameSettings, Random) : GameBoardView
+ stop() : void
- onModelEvent(GameEvent) : void
- showClock() : void
~ roll() : void
- afterRoll(Player, DiceRoll) : void
~ pieceSelected(Piece) : void
- animate(MoveResult, Runnable) : void
- animateCaptures(MoveResult, Runnable) : void
- afterMove(Player, MoveResult) : void
- announce(Player, MoveResult) : String
- startTurn() : void
- startTurn(String, GameBoardView.StatusKind) : void
- finishGame() : void
- later(int, Runnable) : void
+ model() : GameModel
+ view() : GameBoardView
~ isBusy() : boolean

Main

- Main()
+ main(String[]) : void
+ createApp() : NavigationController
- launch() : void

NavigationController

+ MENU : String
+ SETUP : String
+ GAME : String
+ RESULTS : String
- frame : JFrame
- cardLayout : CardLayout
- cards : JPanel
- screens : Map<String, JComponent>
- menuView : MainMenuView
- setupView : GameSetupView
- resultsView : GameResultsView
- game : GameController
- gameView : GameBoardView
- lastSettings : GameSettings
- current : String
- exitHandler : Runnable

+ NavigationController()
- register(String, JComponent) : void
- show(String) : void
- defaultButtonOf(JComponent) : JButton
+ showMainMenu() : void
+ showSetup() : void
+ showHistory() : void
+ openQuestionManagement() : void
+ startGame(GameSettings) : void
+ startGame(GameSettings, Random) : void
- playAgain() : void
+ showResults(List<Player>, Difficulty, String) : void
+ confirmLeaveGame(Runnable) : void
+ requestExit() : void
+ setExitHandler(Runnable) : void
+ overlay(JComponent) : ModalOverlay
+ confirm(String, String, String, String, boolean, Icons.Type, Runnable) : void
+ info(String, String, Icons.Type) : void
+ frame() : JFrame
+ currentScreen() : String
+ game() : GameController
+ menuView() : MainMenuView
+ setupView() : GameSetupView
+ gameView() : GameBoardView
+ resultsView() : GameResultsView
- previewGame() : GameModel
- appIcon(int) : Image

Relationships between the classes

Every relationship between two classes of the system, generated from the compiled code (source as of commit 18b21c5)

generalization = extends · realization = implements an interface · association = held in a field (1, * = collection, 0..1 = Optional) · nesting = inner class of

Class	Relationship	Other class	Through (field [multiplicity])
Board	association	Board.Corner	corners [*]
Board	association	Cell	squares [*], paths [*]
Board	association	Difficulty	difficulty [1]
Board	association	PlayerColor	paths [*], corners [*]
Board	association	Square	squares [*]
Board.Corner	nesting	Board	nested in
BoardComponent	association	BoardComponent.PieceListener	listener [1]
BoardComponent	association	GameModel	state [1]
BoardComponent	association	Piece	targets [*], hovered [1], overrides [*]
BoardComponent.PieceListener	nesting	BoardComponent	nested in
ColorSwatch	association	PlayerColor	color [1]
Confetti	association	Confetti.Particle	particles [*]
Confetti.Particle	nesting	Confetti	nested in
ConfirmDialog	generalization	CustomPanel	extends
ConfirmDialog	association	CustomButton	confirm [1], cancel [1]
ConfirmDialog.IconBadge	nesting	ConfirmDialog	nested in
ConfirmDialog.IconBadge	association	Icons.Type	icon [1]
CustomButton	association	CustomButton.Variant	variant [1]
CustomButton.Variant	nesting	CustomButton	nested in
CustomPanel	association	CustomPanel.Style	style [1]
CustomPanel.Style	nesting	CustomPanel	nested in
DiceComponent	association	DieType	type [1]
DiceRoll	association	DieType	specialType [1]
GameBoardView	generalization	GradientBackground	extends
GameBoardView	realization	ScreenView	implements
GameBoardView	association	BoardComponent	board [1]
GameBoardView	association	CustomButton	roll [1], back [1]
GameBoardView	association	DiceComponent	regularDie [1], specialDie [1]
GameBoardView	association	GameBoardView.TurnCard	turnCard [1]
GameBoardView	association	GameModel	state [1]
GameBoardView	association	UiKit.Pill	levelPill [1]
GameBoardView.LegendIcon	nesting	GameBoardView	nested in
GameBoardView.LegendIcon	association	SquareType	type [1]
GameBoardView.PlayerRow	nesting	GameBoardView	nested in
GameBoardView.PlayerRow	association	PlayerColor	color [1]
GameBoardView.StatusKind	nesting	GameBoardView	nested in
GameBoardView.TurnCard	nesting	GameBoardView	nested in
GameBoardView.TurnCard	association	PlayerColor	color [1]
GameController	association	GameBoardView	view [1]
GameController	association	GameModel	model [1]
GameController	association	NavigationController	nav [1]
GameEvent	association	DiceRoll	roll [1]
GameEvent	association	GameEvent.Type	type [1]
GameEvent	association	MoveResult	move [1]
GameEvent	association	Player	player [1]
GameEvent.Type	nesting	GameEvent	nested in
GameModel	association	Board	board [1]
GameModel	association	GameClock	clock [1]
GameModel	association	GameListener	listeners [*]
GameModel	association	GameSettings	settings [1]
GameModel	association	Player	players [*]
GameModel	association	RankingTracker	ranking [1]
GameModel	association	TurnManager	turns [1]
GameModel.EngineFactory	nesting	GameModel	nested in
GameResultsView	realization	ScreenView	implements
GameResultsView	association	Confetti	confetti [1]
GameResultsView	association	CustomButton	playAgain [1], menu [1]
GameResultsView	association	GameResultsView.Podium	podium [1]
GameResultsView.Podium	nesting	GameResultsView	nested in
GameResultsView.Podium	association	Player	ranking [*]
GameSettings	association	Difficulty	difficulty [1]
GameSettings	association	GameSettings.PlayerSetup	players [*]
GameSettings.FieldError	nesting	GameSettings	nested in
GameSettings.PlayerSetup	nesting	GameSettings	nested in
GameSettings.PlayerSetup	association	PlayerColor	color [1]
GameSetupView	generalization	GradientBackground	extends
GameSetupView	realization	ScreenView	implements
GameSetupView	association	CustomButton	back [1], start [1]
GameSetupView	association	Difficulty	difficultyCards [*], difficulty [1], startHandler [*]
GameSetupView	association	GameSettings.FieldError	startHandler [*]
GameSetupView	association	GameSettings.PlayerSetup	startHandler [*]
GameSetupView	association	GameSetupView.DifficultyCard	difficultyCards [*]
GameSetupView	association	GameSetupView.PlayerRow	rows [*]
GameSetupView	association	PlayerColor	DEFAULT_COLORS [*]
GameSetupView	association	ToggleChip	countChips [*]
GameSetupView.DifficultyCard	nesting	GameSetupView	nested in
GameSetupView.DifficultyCard	association	Difficulty	level [1]
GameSetupView.PlayerRow	nesting	GameSetupView	nested in
GameSetupView.PlayerRow	association	ColorSwatch	swatches [*]
GameSetupView.PlayerRow	association	GameSetupView.PlayerRow.Badge	badge [1]
GameSetupView.PlayerRow	association	PlayerColor	color [1]
GameSetupView.PlayerRow	association	StyledTextField	name [1]
GameSetupView.PlayerRow.Badge	nesting	GameSetupView.PlayerRow	nested in
Icons.Type	nesting	Icons	nested in
Icons.VectorIcon	nesting	Icons	nested in
Icons.VectorIcon	association	Icons.Type	type [1]
LevelConfig	association	DieType	specialDie [1]
LevelConfig	association	Difficulty	LEVELS [*]
MainMenuView	generalization	GradientBackground	extends
MainMenuView	realization	ScreenView	implements
MainMenuView	association	CustomButton	newGame [1], history [1], questions [1], exit [1]
MainMenuView	association	MainMenuView.Hero	hero [1]
MainMenuView.ComingSoonTag	nesting	MainMenuView	nested in
MainMenuView.Hero	nesting	MainMenuView	nested in
MainMenuView.Hero	association	BoardComponent	board [1]
MainMenuView.Hero	association	DiceComponent	die [1], boost [1]
MainMenuView.Option	nesting	MainMenuView	nested in
MainMenuView.Option	association	MainMenuView.ComingSoonTag	tag [1]
MainMenuView.Title	nesting	MainMenuView	nested in
Move	association	Piece	piece [1]
MoveResult	association	Cell	cells [*], fromCell [1]
MoveResult	association	MoveResult.Capture	captures [*]
MoveResult	association	MoveResult.Kind	kind [1]
MoveResult	association	Piece	piece [1]
MoveResult	association	Square	landedOn [1]
MoveResult.Capture	nesting	MoveResult	nested in
MoveResult.Capture	association	Cell	from [1]
MoveResult.Capture	association	Piece	piece [1]
MoveResult.Kind	nesting	MoveResult	nested in
NavigationController	association	GameBoardView	gameView [1]
NavigationController	association	GameController	game [1]
NavigationController	association	GameResultsView	resultsView [1]
NavigationController	association	GameSettings	lastSettings [1]
NavigationController	association	GameSetupView	setupView [1]
NavigationController	association	MainMenuView	menuView [1]
Piece	association	PlayerColor	color [1]
Player	association	Piece	pieces [*]
Player	association	PlayerColor	color [1]
RandomDice	realization	Dice	implements
Square	association	Cell	cell [1]
Square	association	Difficulty	questionLevel [1]
Square	association	Direction	arrowDirection [1]
Square	association	PlayerColor	owner [1]
Square	association	SquareType	type [1]
StandardHomePathRules	realization	HomePathRules	implements
StandardHomePathRules	association	Board	board [1]
StandardMoveEngine	realization	MoveEngine	implements
StandardMoveEngine	association	Board	board [1]
StandardMoveEngine	association	HomePathRules	homePathRules [1]
StandardMoveEngine	association	Move	offered [*]
StandardMoveEngine	association	Piece	offered [*]
StandardMoveEngine	association	Player	players [*]
StandardMoveEngine	association	SquareRules	squareRules [1]
StandardRankingTracker	realization	RankingTracker	implements
StandardRankingTracker	association	Player	players [*], finished [*]
StandardSquareRules	realization	SquareRules	implements
StandardSquareRules	association	Board	board [1]
StandardSquareRules	association	Player	players [*]
StandardTurnManager	realization	TurnManager	implements
StandardTurnManager	association	Dice	dice [1]
StandardTurnManager	association	DiceRoll	lastRoll [1]
StandardTurnManager	association	Difficulty	level [1]
StandardTurnManager	association	Move	cachedMoves [*]
StandardTurnManager	association	MoveEngine	moveEngine [1]
StandardTurnManager	association	Piece	cachedMoves [*]
StandardTurnManager	association	Player	players [*]
StandardTurnManager	association	RankingTracker	ranking [1]
StandardTurnManager	association	TurnPhase	phase [1]
StyledTable.HeaderRenderer	nesting	StyledTable	nested in
StyledTable.PillRenderer	nesting	StyledTable	nested in
StyledTable.PillRenderer	association	Difficulty	level [1]
StyledTable.TextRenderer	nesting	StyledTable	nested in
StyledTextField	association	StyledTextField.FieldLook	look [1]
StyledTextField.FieldLook	nesting	StyledTextField	nested in
StyledTextField.Password	nesting	StyledTextField	nested in
StyledTextField.Password	association	StyledTextField.FieldLook	look [1]
UiKit.Pill	nesting	UiKit	nested in